PES-VCS: A Simplified Version Control System Implementation

Name : Shrikar Desai

SRN : PES1UG24CS916

Screenshot 1A: Phase 1 Object Storage Test Output

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Screenshot 1B: Stored Object Files in .pes/objects

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

Screenshot 2A: Phase 2 Tree Serialization Test Output

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Screenshot 2B: Commit Object Hex Dump Showing Tree Hash and Metadata

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ xxd .pes/objects/ea/c283dd874e5863d728ab56363280e633c7f8b3f398aba028b77a7c117647c1 | head -20
00000000: 636f 6d6d 6974 2031 3831 0074 7265 6520  commit 181.tree
00000010: 3665 6631 3962 3431 3232 3563 3533 3639  6ef19b41225c5369
00000020: 6631 6331 3034 6434 3564 3864 3835 6566  f1c104d45d8d85ef
00000030: 6139 6230 3537 6235 3362 3134 6234 6239  a9b057b53b14b4b9
00000040: 6239 3339 6464 3734 6465 6363 3533 3231  b939dd74decc5321
00000050: 0a61 7574 686f 7220 5368 7269 6b61 7220  .author Shrikar
00000060: 4465 7361 6920 3c50 4553 3155 4732 3443  Desai <PES1UG24C
00000070: 5339 3136 3e20 3137 3736 3431 3339 3230  S916> 1776413920
00000080: 0a63 6f6d 6d69 7474 6572 2053 6872 696b  .committer Shrik
00000090: 6172 2044 6573 6169 203c 5045 5331 5547  ar Desai <PES1UG
000000a0: 3234 4353 3931 363e 2031 3737 3634 3133  24CS916> 1776413
000000b0: 3932 300a 0a43 7265 6174 6520 7472 6565  920..Create tree
```

Screenshot 3A: Staging Files and Checking Status

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ make pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto

┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ rm -rf .pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt
./pes add file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  compat.h
  untracked:  .gitignore
  untracked:  Makefile
  untracked:  commit.c
  untracked:  hello.txt
  untracked:  tree.c
  untracked:  README.md
  untracked:  pes.h
  untracked:  bye.txt
  untracked:  index.h
  untracked:  pes.c
  untracked:  file.txt
  untracked:  test_sequence.sh
  untracked:  index.c
  untracked:  commit.h
  untracked:  object.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  tree.h


┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$
```

Screenshot 3B: Contents of .pes/index After Staging Files

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776428898 6 file1.txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776428898 6 file2.txt
```

Screenshot 4A: Commit History Using ./pes log

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ rm -rf .pes
./pes init

echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 491d4dc94f12 ... Initial commit
Committed: ce0fc44daf20 ... Add world
Committed: 407e4915fedf ... Add farewell
commit 407e4915fedf53ec7df603e7f512802b7b0a65eebf99e6e86aaa1fffc461cb21
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429146

    Add farewell

commit ce0fc44daf206d4cbdf95768b928df477009587cea273d6e9fc686a58d80f38c
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429146

    Add world

commit 491d4dc94f12023c9a6c812bcb2efbdda015144bc0b9a5ccb39d3035e7b8e741
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429146

    Initial commit
```

Screenshot 4B: Repository File Structure After Multiple Commits

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/40/7e4915fedf53ec7df603e7f512802b7b0a65eebf99e6e86aaa1fffc461cb21
.pes/objects/49/1d4dc94f12023c9a6c812bcb2efbdda015144bc0b9a5ccb39d3035e7b8e741
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/ce/0fc44daf206d4cbdf95768b928df477009587cea273d6e9fc686a58d80f38c
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

Screenshot 4C: HEAD and Main Branch Reference Contents

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ cat .pes/refs/heads/main
cat .pes/HEAD
407e4915fedf53ec7df603e7f512802b7b0a65eebf99e6e86aaa1fffc461cb21
ref: refs/heads/main
```

Final Integration Test :

```
┌──(kali㉿kali)-[~/Desktop/PES1UG24CS916-pes-vcs]
└─$ make test-integration
═══ Running integration tests ═══
bash test_sequence.sh
═══ PES-VCS Integration Test ═══

── Repository Initialization ──
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

── Staging Files ──
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


── First Commit ──
Committed: a67b18e60f7e... Initial commit

Log after first commit:
commit a67b18e60f7eed86e2e8cc0505dcc15e09fe54c7c0bee4bcfff3a25c90a46ded
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429286

    Initial commit


── Second Commit ──
Committed: 28770a7705fd... Update file.txt

── Third Commit ──
Committed: 1f9175f18518... Add farewell
```

```
── Full History ──
commit 1f9175f185189a66a23bf0656e3e7352fa9bfc238b05583f6568a08d0cd49eb8
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429286

    Add farewell

commit 28770a7705fd4697ee443034e927471427095935ecf8eaa14caad768c3758bd0
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429286

    Update file.txt7◆◆U

commit a67b18e60f7eed86e2e8cc0505dcc15e09fe54c7c0bee4bcfff3a25c90a46ded
Author: Shrikar Desai <PES1UG24CS916>
Date:   1776429286

    Initial commit


── Reference Chain ──
HEAD:
ref: refs/heads/main
refs/heads/main:
1f9175f185189a66a23bf0656e3e7352fa9bfc238b05583f6568a08d0cd49eb8

── Object Store ──
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/1f/9175f185189a66a23bf0656e3e7352fa9bfc238b05583f6568a08d0cd49eb8
.pes/objects/28/770a7705fd4697ee443034e927471427095935ecf8eaa14caad768c3758bd0
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/a6/7b18e60f7eed86e2e8cc0505dcc15e09fe54c7c0bee4bcfff3a25c90a46ded
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

═══ All integration tests completed ═══
```